Project Report on

Time Series Modeling of S&P 500 Returns

Submitted in fulfilment of requirement for the SI509 : Time Series Analysis Course Project

MSc. Applied Statistics and Informatics

Of

Indian Institute of Technology (IIT) Bombay

By

Kamesh Dubey 22N0088

INDIAN INSTITUTE OF TECHNOLOGY
BOMBAY

Under the guidance of

Prof. Sanjeev V Sabnis

Department of Mathematics

Spring 2024

Submission Date: 28 April 2024

INDIAN INSTITUTE OF TECHNOLOGY (IIT) BOMBAY

Powai, Mumbai, Maharashtra 400076

Contents

1 Introduction **3**
1.1 Why SPY? . . . 3

2 Data and Preprocessing **3**
2.1 Data Download . . . 3
2.1.1 Daily Return Sanity Check . . . 4
2.2 Log Returns . . . 4
2.3 Monthly Aggregation . . . 4
2.4 After Preprocessing . . . 4

3 Exploratory Data Analysis **5**
3.1 Summary Statistics . . . 5
3.2 Price Series . . . 5
3.3 Monthly Log Returns . . . 6
3.4 Histogram + KDE . . . 6
3.5 Boxplot . . . 7
3.6 Rolling 12-Month Mean . . . 7
3.7 Rolling 12-Month Volatility . . . 8
3.8 Squared Returns . . . 8

4 Stationarity and Dependence Tests **9**
4.1 ADF Test . . . 9
4.2 KPSS Test . . . 9
4.3 Combined Verdict . . . 9
4.4 ACF of Returns . . . 10
4.5 PACF of Returns . . . 10

5 Mean Modeling: ARIMA **10**
5.1 Grid Search . . . 10
5.2 Selected: ARIMA(0,0,0) . . . 11

6 Mean Model Diagnostics **12**
6.1 Residual Plots . . . 12
6.2 Ljung-Box Test . . . 14
6.3 Jarque-Bera Test . . . 14

7 Testing for ARCH Effects **15**
7.1 ARCH-LM Test 15
7.2 ACF of Squared Residuals 15

8 Volatility Modeling: ARCH/GARCH **15**
8.1 Candidates 15
8.2 Comparison 16
8.3 Selected: GARCH(1,1) 16

9 Volatility Model Diagnostics **17**
9.1 Conditional Volatility 17
9.2 Standardized Residuals 18
9.3 ACF of Squared Standardized Residuals 20
9.4 Ljung-Box Tests 20

10 Additional Analysis **21**
10.1 Rolling vs Fitted Volatility 21
10.2 Volatility Forecast 21

11 Conclusions and Limitations **22**
11.1 Findings 22
11.2 Limitations 22
11.3 Future Work 22

A Software **23**

1 Introduction

We analyze monthly log returns of **SPY** (S&P 500 ETF) from Jan 2006 to Dec 2022. 204 monthly observations covering two major crises (2008 GFC and 2020 COVID). The goal is to fit an ARIMA mean model and a GARCH volatility model using a data-driven workflow:

1. Data download and quality checks
2. Preprocessing: monthly resampling, log returns
3. EDA: summary stats, distribution, rolling stats
4. Stationarity tests (ADF, KPSS) and ACF/PACF
5. ARIMA mean model selection via AIC/BIC
6. ARCH-LM test for conditional heteroskedasticity
7. GARCH volatility model selection via AIC/BIC
8. Diagnostics on standardized residuals

All model choices are data-driven. We compare candidates and let information criteria decide.

1.1 Why SPY?

SPY tracks the S&P 500 index. Liquid, clean data, no gaps. Monthly returns show the standard stylized facts: weak autocorrelation, volatility clustering, fat tails.

2 Data and Preprocessing

2.1 Data Download

Downloaded daily SPY prices from Yahoo Finance using `yfinance`. Range: Dec 2005 to Feb 2023 (padded for clean month-end resampling).

Table 1: Download summary

Ticker	SPY
Range	2005-12-01 to 2023-01-31
Daily rows	4 320
Columns	Close, High, Low, Open, Volume
Missing	0
Duplicates	0

Yahoo's API puts split-adjusted prices into `Close` directly. Zero missing, zero duplicates, sorted chronologically.

2.1.1 Daily Return Sanity Check

Table 2: Daily return stats

Daily returns	4 319
Mean (daily)	0.000427
Std (daily)	0.012573
Min	−0.1094
Max	0.1452
Annualised return ($\times 252$)	$\approx$ 10.8%
Annualised vol ($\times\sqrt{252}$)	$\approx$ 20.0%

Annualised return $\approx$ 11% and vol $\approx$ 20%. Includes the 2008 crisis period. Min −10.9% is COVID crash, max +14.5% is a crisis rebound day.

2.2 Log Returns

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Additive over time: $r_{t_1 \to t_n} = \sum r_i$.

2.3 Monthly Aggregation

Resample to month-end, compute log returns. Monthly data reduces microstructure noise. Tradeoff: $n = 204$, still limited for GARCH but much better than ∼100 obs from a shorter window.

2.4 After Preprocessing

- Date range: Jan 2006 to Dec 2022
- Monthly observations: **204**
- Missing values: 0
- Price column: Close

3 Exploratory Data Analysis

3.1 Summary Statistics

Table 3: Monthly log return summary

Mean	0.0070
Median	0.0135
Std	0.0453
Min	−0.1805
Max	0.1195
Skewness	−0.79
Excess Kurtosis	1.51

Mean $\approx$ 0.7%/month. Median higher (1.35%), negative skewness pulling mean down. Excess kurtosis = 1.51, heavier tails than $\mathcal{N}$. Min = −18.1% is Oct 2008 (GFC), max $\approx$ +12%.

3.2 Price Series

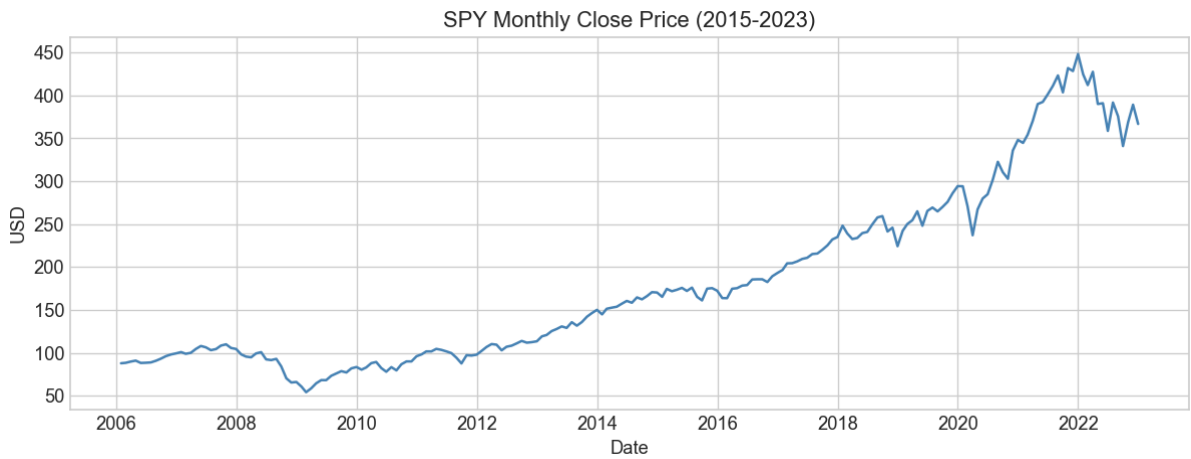

Figure 1: SPY monthly close price (2006–2022)

~\$65 (2006) to ~\$370 (2022). Three dips: 2008 GFC, COVID 2020, 2022 drawdown. Non-stationary, thats why we model returns.

3.3 Monthly Log Returns

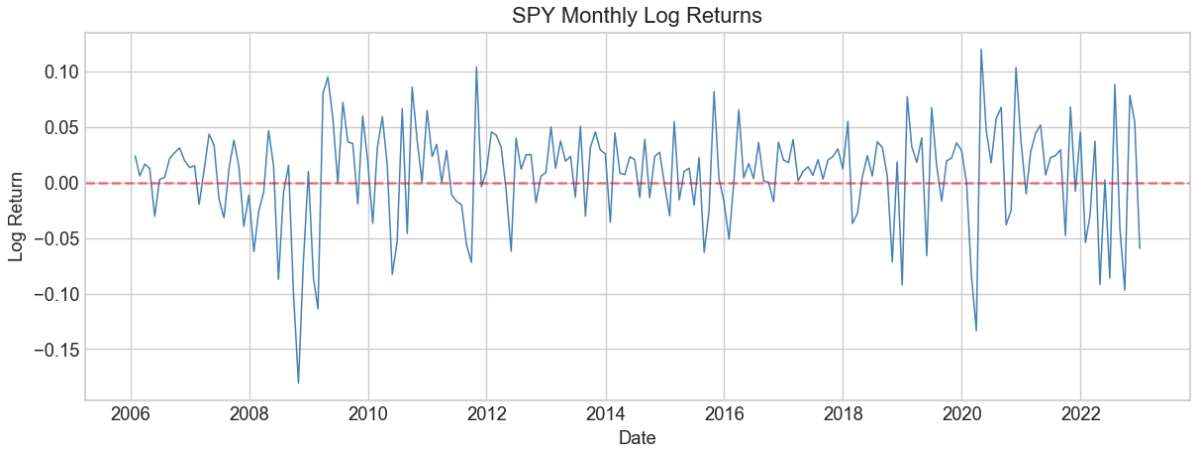

Figure 2: SPY monthly log returns (2006–2022)

Fluctuates around zero. Amplitude not constant: big swings in 2008-09, 2011, 2020, 2022. Volatility clustering: large $|r_t|$ followed by large $|r_{t+1}|$.

3.4 Histogram + KDE

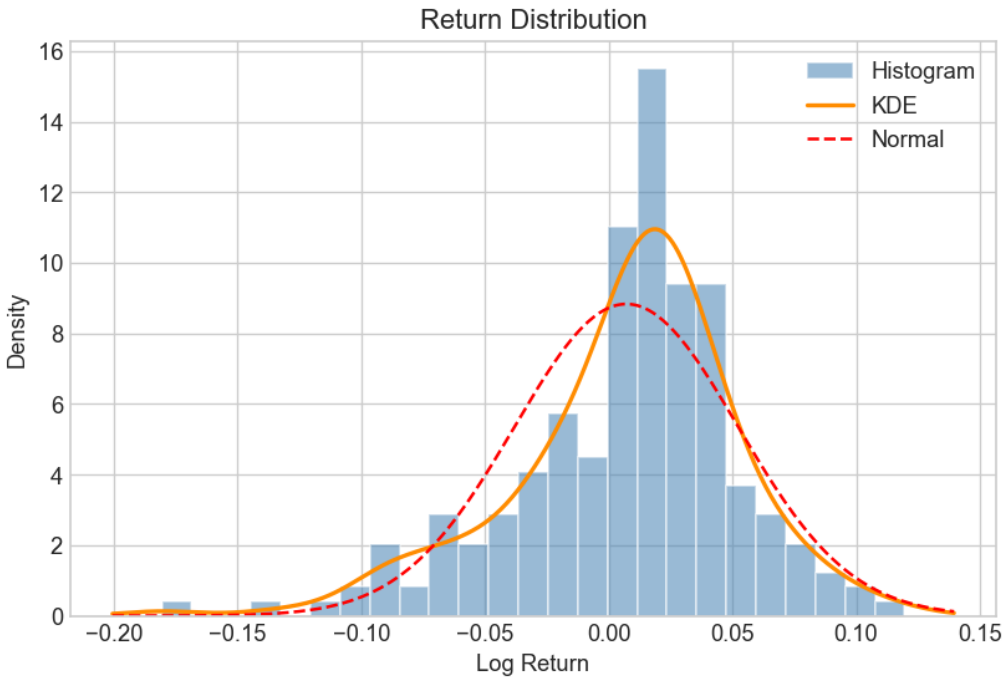

Figure 3: SPY monthly log return distribution

KDE has fatter left tail than normal. Skewness $= -0.79$, leptokurtic (excess kurtosis $= 1.51$).

3.5 Boxplot

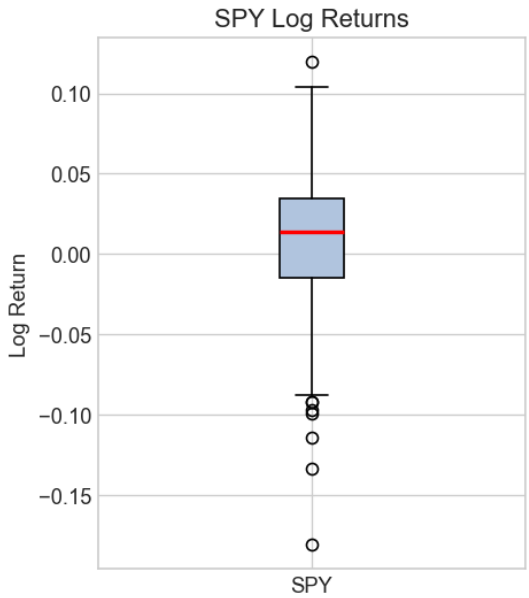

Figure 4: Boxplot of SPY monthly log returns

Median above zero. Extreme negative outlier is Oct 2008.

3.6 Rolling 12-Month Mean

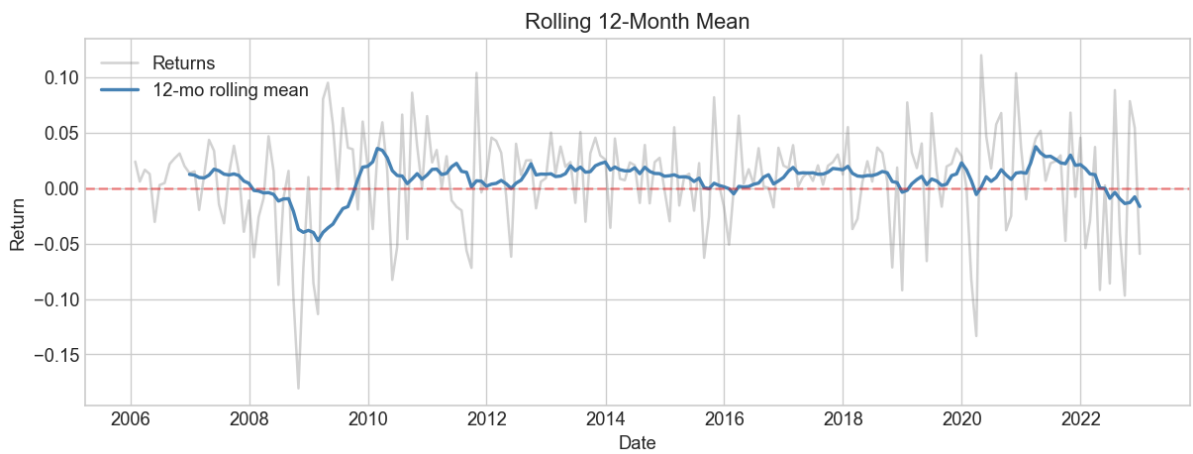

Figure 5: Rolling 12-month mean of SPY monthly log returns

Stays near zero, dips during downturns (2008-09, 2011, 2020). Consistent with stationarity.

3.7 Rolling 12-Month Volatility

Figure 6: Rolling 12-month volatility of SPY monthly log returns

Spikes in 2008-09 (GFC) and 2020 (COVID). $\mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$ is not constant. Textbook volatility clustering.

3.8 Squared Returns

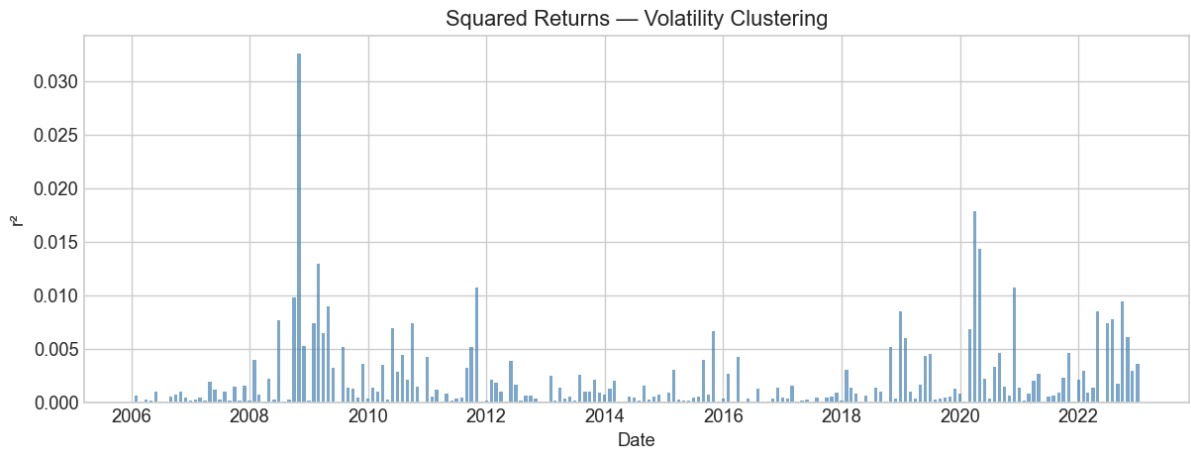

Figure 7: Squared monthly log returns r_t^2 of SPY

r_t^2 clusters together: 2008-09, 2011, 2018-Q4, 2020-Q1, 2022.

$$E\left[r_t^2 \mid r_{t-1}^2\right] \neq E\left[r_t^2\right]$$

ARCH effects likely present.

4 Stationarity and Dependence Tests

4.1 ADF Test

H_0: unit root. H_1: stationary. Reject if $p < 0.05$.

Table 4: ADF results

ADF statistic	-5.9862
p-value	< 0.05
Lags used	5
Critical 1%	-3.4640
Critical 5%	-2.8763

$p < 0.05$, reject H_0. Series is stationary.

4.2 KPSS Test

H_0: stationary. H_1: unit root. Fail to reject if $p > 0.05$.

Table 5: KPSS results

KPSS statistic	0.1379
p-value	> 0.05
Critical 5%	0.4630

$p > 0.05$, fail to reject H_0. Stationary.

4.3 Combined Verdict

Both ADF and KPSS agree $\Rightarrow$ returns are stationary, $I(0)$. Use $d = 0$ for ARIMA.

4.4 ACF of Returns

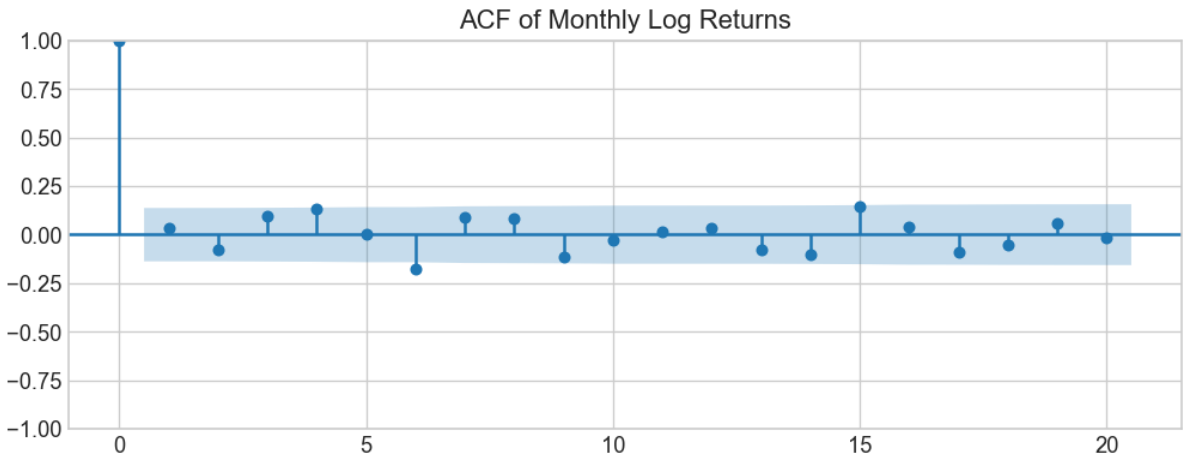

Figure 8: ACF of SPY monthly log returns

Almost all lags inside 95% band. $\hat{\rho}(h) \approx 0$ for $h \geq 1$. Weak autocorrelation.

4.5 PACF of Returns

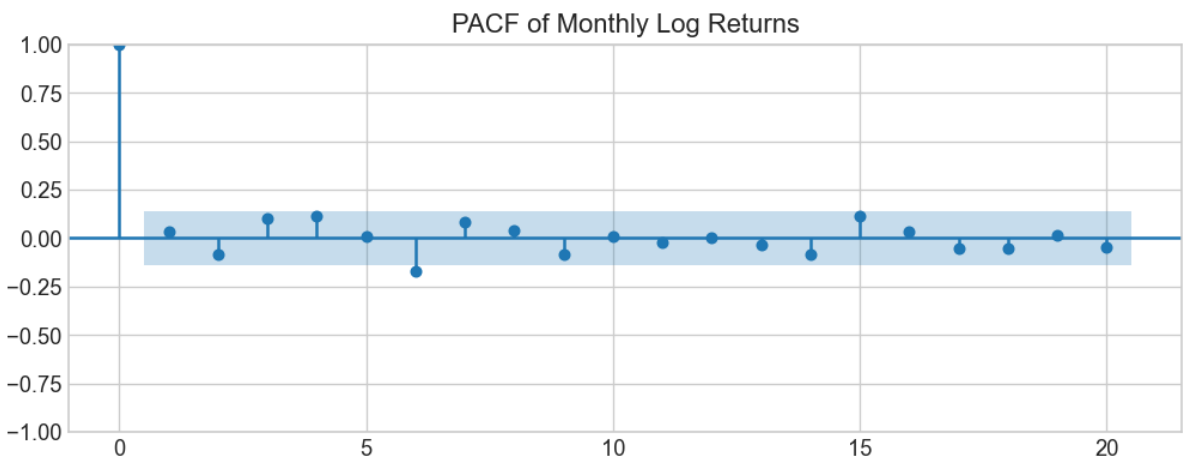

Figure 9: PACF of SPY monthly log returns

PACF clean. Suggests ARIMA(0,0,0): $r_t = \mu + \varepsilon_t$.

5 Mean Modeling: ARIMA

5.1 Grid Search

Fit ARIMA(p, 0, q) for $p, q \in \{0, 1, 2\}$, 9 models. All converged. Rank by BIC.

ARIMA(0,0,0) has lowest BIC ($= -675.12$). BIC differences across candidates are small, mean structure is very weak.

5.2 Selected: ARIMA(0,0,0)

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t \sim \mathrm{WN}(0, \sigma^2)$$

ARIMA(0,0,0) Model Summary

```
                               SARIMAX Results
==============================================================================
Dep. Variable:             Log_Return   No. Observations:                  204
Model:                          ARIMA   Log Likelihood                 342.878
Date:                Mon, 30 Mar 2026   AIC                           -681.757
Time:                        03:25:24   BIC                           -675.120
Sample:                    01-31-2006   HQIC                          -679.072
                         - 12-31-2022
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0071      0.003      2.044      0.041       0.000       0.014
sigma2         0.0020      0.000     12.041      0.000       0.002       0.002
===================================================================================
Ljung-Box (L1) (Q):                   0.26   Jarque-Bera (JB):                39.35
Prob(Q):                              0.61   Prob(JB):                         0.00
Heteroskedasticity (H):               1.02   Skew:                            -0.79
Prob(H) (two-sided):                  0.94   Kurtosis:                         4.46
===================================================================================
```

Figure 10: ARIMA(0,0,0) model summary

$\hat{\mu} = 0.0071$ ($\approx 0.7\%$/month), $p = 0.041 < 0.05$, significant. $\hat{\sigma}^2 = 0.0020$. JB rejects normality ($p < 0.05$, skewness $= -0.79$, kurtosis $= 4.46$).

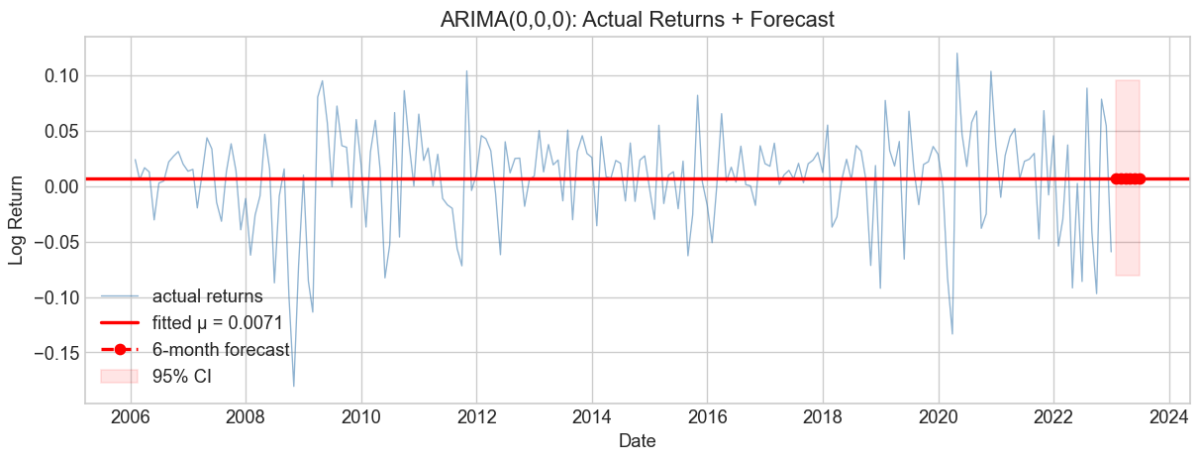

Figure 11: ARIMA(0,0,0): actual returns + 6-month forecast with 95% CI

Forecast is flat at $\hat{\mu}$ since ARIMA(0,0,0) has no AR/MA dynamics. The 95% CI band shows the uncertainty. Most returns fall within it but extreme months (2008, 2020) blow past it.

6 Mean Model Diagnostics

If correct, residuals $\hat{\varepsilon}_t = r_t - \hat{\mu}$ should be white noise.

6.1 Residual Plots

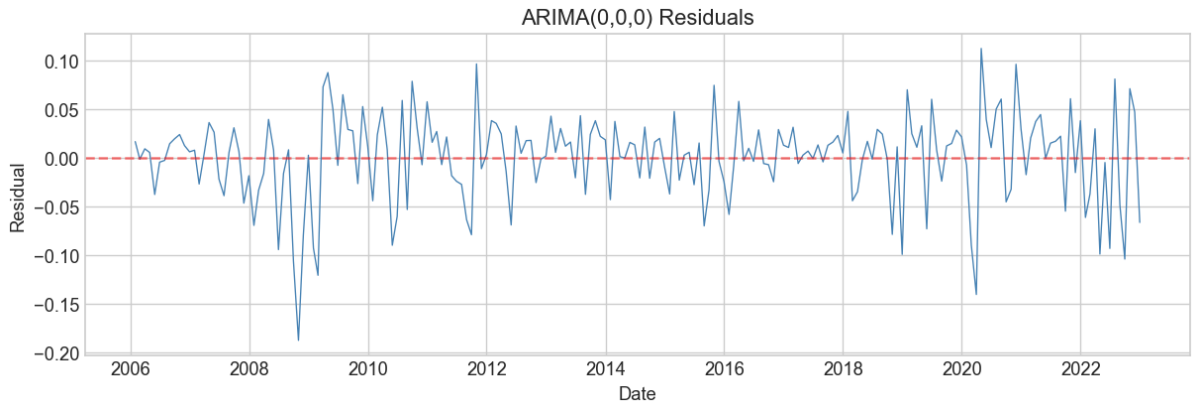

Figure 12: ARIMA(0,0,0) residuals over time

No trend. Variance not constant, big swings in 2008-09 and 2020. GARCH will handle this.

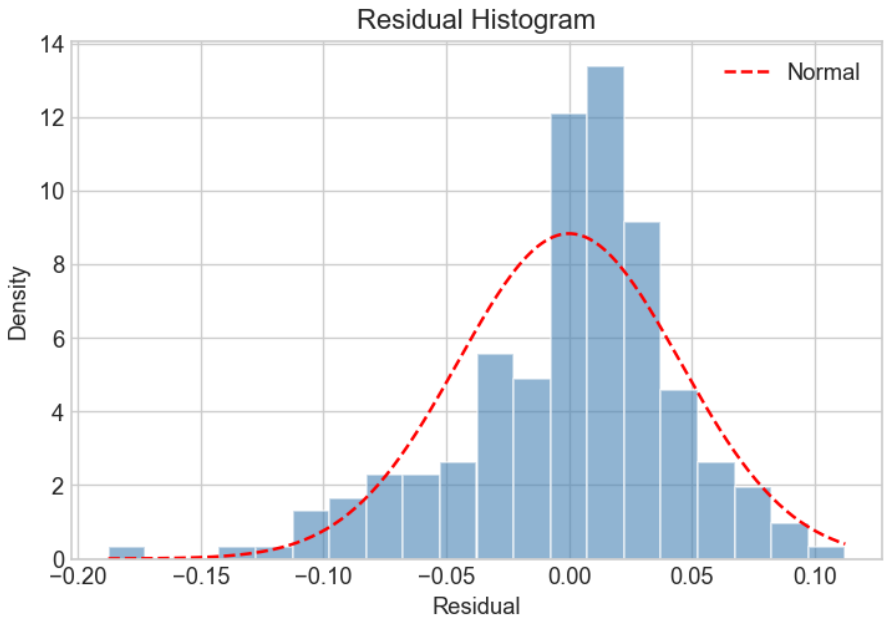

Figure 13: ARIMA(0,0,0) residual histogram with normal overlay

Heavier tails than normal. Kurtosis $= 4.46 > 3$ (leptokurtic), skewness $= -0.79$.

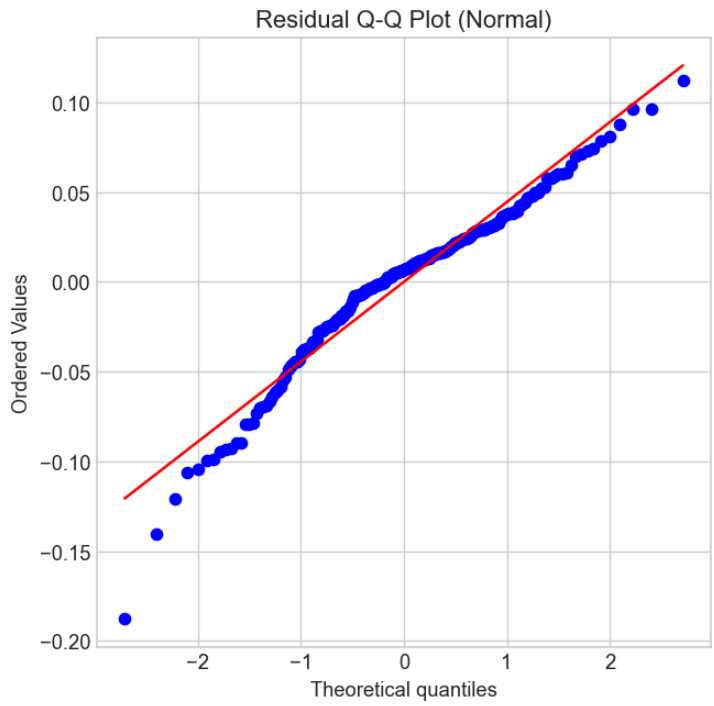

Figure 14: Q-Q plot of ARIMA(0,0,0) residuals vs normal

Tails deviate, especially lower left. Confirms JB rejection. MLE still consistent.

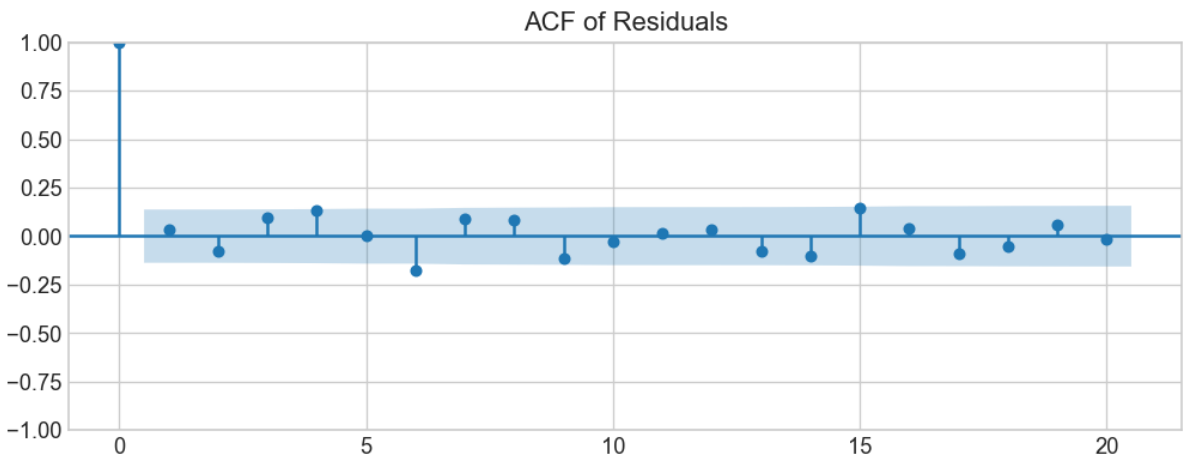

Figure 15: ACF of ARIMA(0,0,0) residuals

All lags within 95% band. No autocorrelation left.

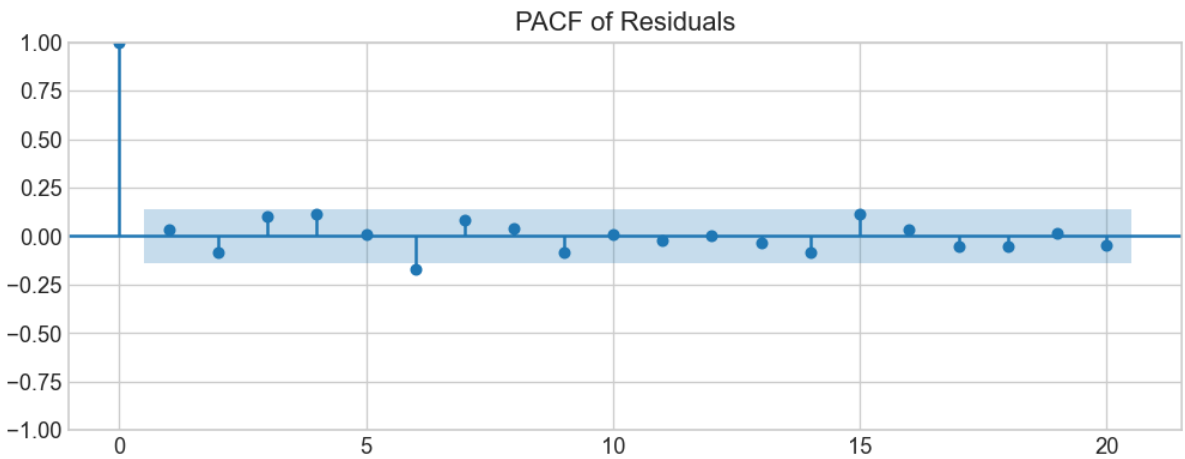

Figure 16: PACF of ARIMA(0,0,0) residuals

PACF clean. No hidden AR structure.

6.2 Ljung-Box Test

$$Q(m) = n(n+2)\sum_{k=1}^{m}\frac{\hat{\rho}_k^2}{n-k} \sim \chi^2(m)$$

H_0: no autocorrelation. Reject if $p < 0.05$.

Table 6: Ljung-Box on residuals

Lag	LB stat	p-value	Result
5	7.06	0.22	no autocorr
10	20.11	0.03	borderline
15	28.80	0.02	borderline
20	32.68	0.04	borderline

Lag 5: $p = 0.22$, no autocorrelation. Lags 10+: $p < 0.05$, some weak autocorrelation at longer lags. Borderline, not unusual with 204 obs. Mean model is still adequate.

6.3 Jarque-Bera Test

$$JB = \frac{n}{6}\left(S^2 + \frac{(K-3)^2}{4}\right) \sim \chi^2(2)$$

H_0: normal. Reject if $p < 0.05$.

JB = 39.35, $p < 0.05$, reject H_0. Residuals not normal. Skewness = -0.79, kurtosis = 4.46. Doesnt affect ARIMA consistency.

7 Testing for ARCH Effects

7.1 ARCH-LM Test

$$\hat{\varepsilon}_t^2 = \alpha_0 + \alpha_1 \hat{\varepsilon}_{t-1}^2 + \cdots + \alpha_q \hat{\varepsilon}_{t-q}^2 + u_t$$

H_0: no ARCH effects. Reject if $p < 0.05$.

Table 7: ARCH-LM test

Lags	LM stat	p-value	Result
1	25.69	< 0.05	ARCH effects
4	36.44	< 0.05	ARCH effects
8	35.71	< 0.05	ARCH effects
12	37.03	< 0.05	ARCH effects

Reject H_0 at all four lags. ARCH effects clearly present. Need GARCH.

7.2 ACF of Squared Residuals

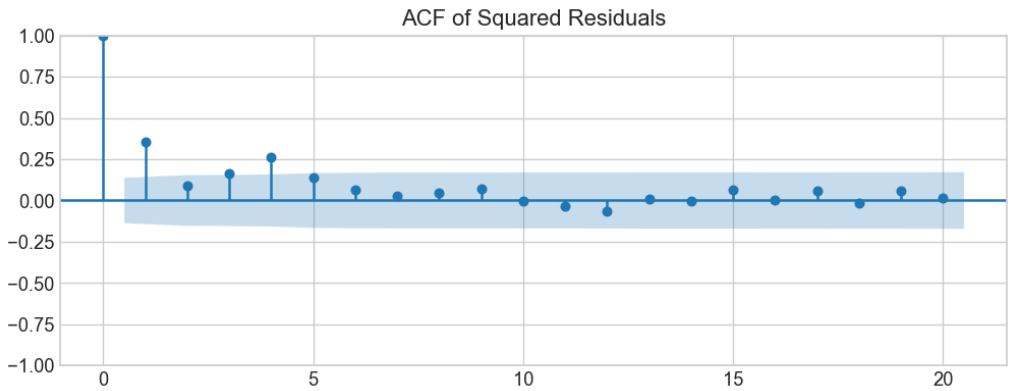

Figure 17: ACF of squared ARIMA residuals $\hat{\varepsilon}_t^2$

Significant spikes at early lags. $\hat{\varepsilon}_t^2$ is autocorrelated, variance is predictable.

8 Volatility Modeling: ARCH/GARCH

8.1 Candidates

ARCH(q) for $q \in \{1, 2, 3\}$ and GARCH(p,q) for $p, q \in \{1, 2\}$. Returns scaled by 100 for numerical stability.

8.2 Comparison

Table 8: Volatility model comparison (by BIC)

Model	AIC	BIC	Log-Lik
GARCH(1,1)	1153.78	1167.03	−572.89
GARCH(2,1)	1155.48	1172.04	−572.74
GARCH(1,2)	1155.78	1172.35	−572.89
GARCH(2,2)	1157.48	1177.35	−572.74
ARCH(1)	1169.44	1179.38	−581.72
ARCH(2)	1169.44	1179.38	−581.72
ARCH(3)	1169.44	1179.38	−581.72

GARCH(1,1) lowest BIC. All ARCH models gave identical AIC/BIC. The β_1 persistence term makes the difference.

8.3 Selected: GARCH(1,1)

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

GARCH(1,1) Model Summary

```
                           AR - GARCH Model Results
==============================================================================
Dep. Variable:             Log_Return   R-squared:                       0.000
Mean Model:                        AR   Adj. R-squared:                  0.000
Vol Model:                      GARCH   Log-Likelihood:               -574.918
Distribution:                  Normal   AIC:                           1157.84
Method:            Maximum Likelihood   BIC:                           1171.11
                                        No. Observations:                  204
Date:                Mon, Mar 30 2026   Df Residuals:                      203
Time:                        03:25:38   Df Model:                            1
                                Mean Model
=====================================================================
                 coef    std err          t      P>|t|  95.0% Conf. Int.
---------------------------------------------------------------------
Const          0.9183      0.226      4.069  4.730e-05 [  0.476,  1.361]
                              Volatility Model
==========================================================================
                 coef    std err          t      P>|t|    95.0% Conf. Int.
--------------------------------------------------------------------------
omega          1.8365      1.324      1.387      0.165   [ -0.759,  4.432]
alpha[1]       0.2871  9.933e-02      2.890  3.847e-03 [9.243e-02,  0.482]
beta[1]        0.6506      0.125      5.214  1.847e-07   [  0.406,  0.895]
==========================================================================

Covariance estimator: robust
```

Figure 18: GARCH(1,1) model summary

α_1 and β_1 both significant ($p < 0.05$). Persistence $= \alpha_1 + \beta_1 = 0.934 < 1$, variance is stationary. Half-life:

$$\frac{\ln 2}{-\ln(0.934)} \approx 10.1 \text{ months}$$

Long-run $\sigma = \sqrt{\omega/(1-\alpha_1-\beta_1)} \approx 5.38\%$ per month.

9 Volatility Model Diagnostics

If correct, $z_t = \hat{\varepsilon}_t / \hat{\sigma}_t$ should be i.i.d. $\sim N(0, 1)$.

9.1 Conditional Volatility

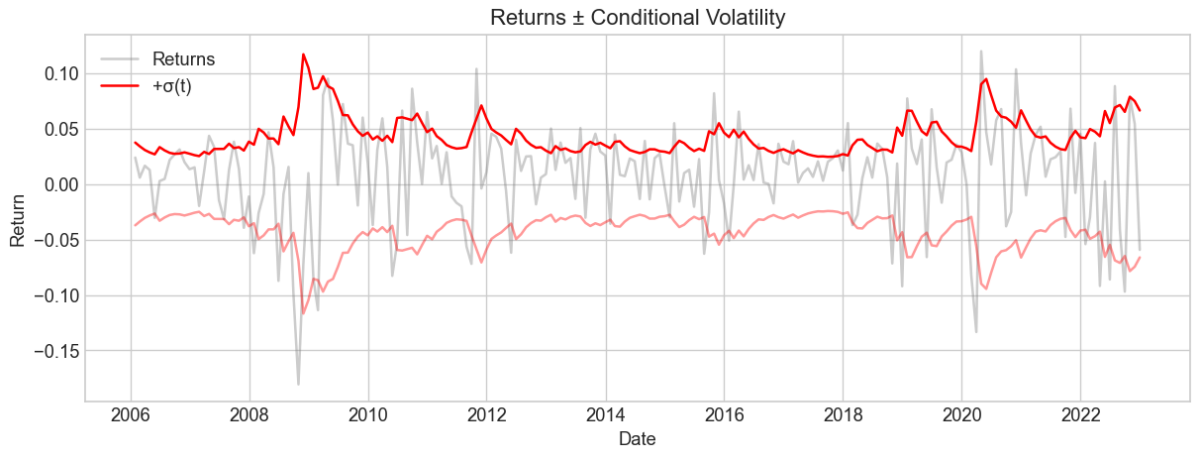

Figure 19: Returns $\pm\hat{\sigma}_t$

Bands widen in 2008-09, 2011, 2020, 2022. Tighten during calm periods. Model adapts.

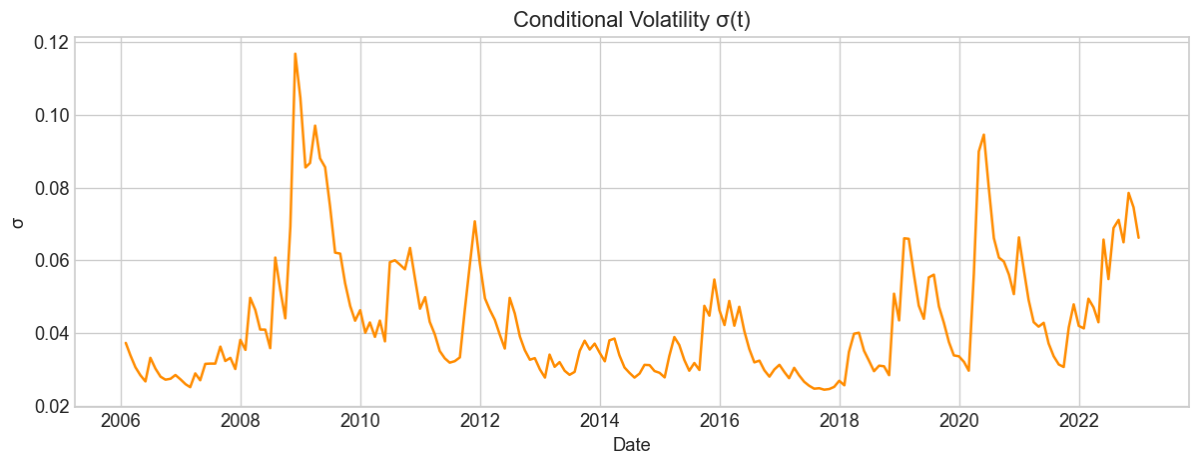

Figure 20: $\hat{\sigma}_t$ over time

Peaks in late 2008, decays slowly (persistence = 0.934). Mean-reverts toward $\bar{\sigma} \approx 5.4\%$.

9.2 Standardized Residuals

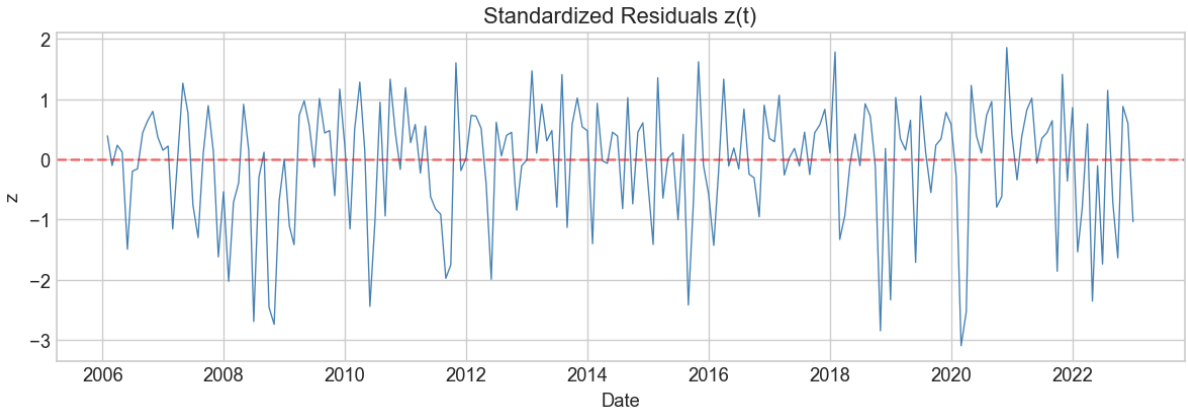

Figure 21: z_t series

No clustering in z_t. GARCH absorbed the time-varying variance.

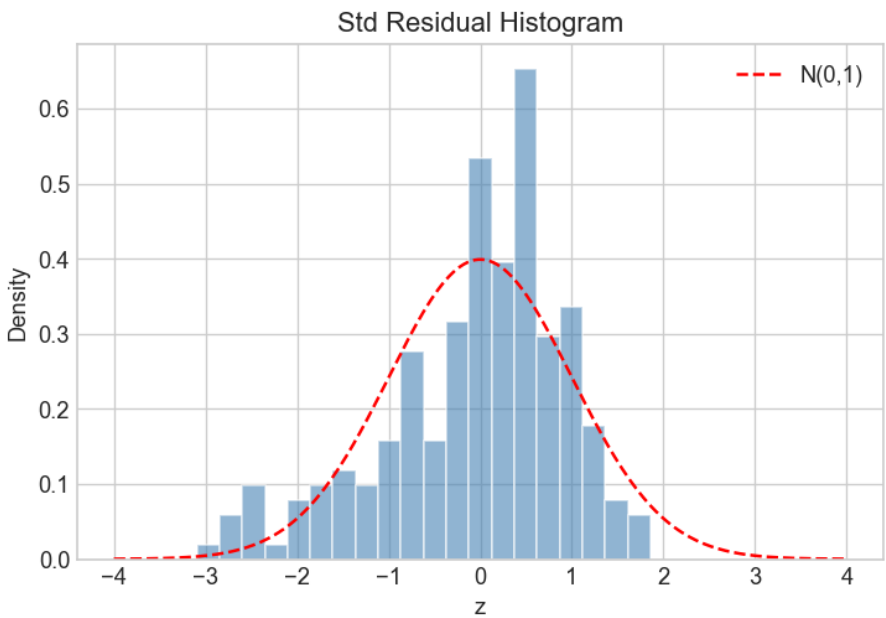

Figure 22: z_t histogram vs $N(0, 1)$

Roughly matches $N(0, 1)$. Slight heavy tails, Student-t would help.

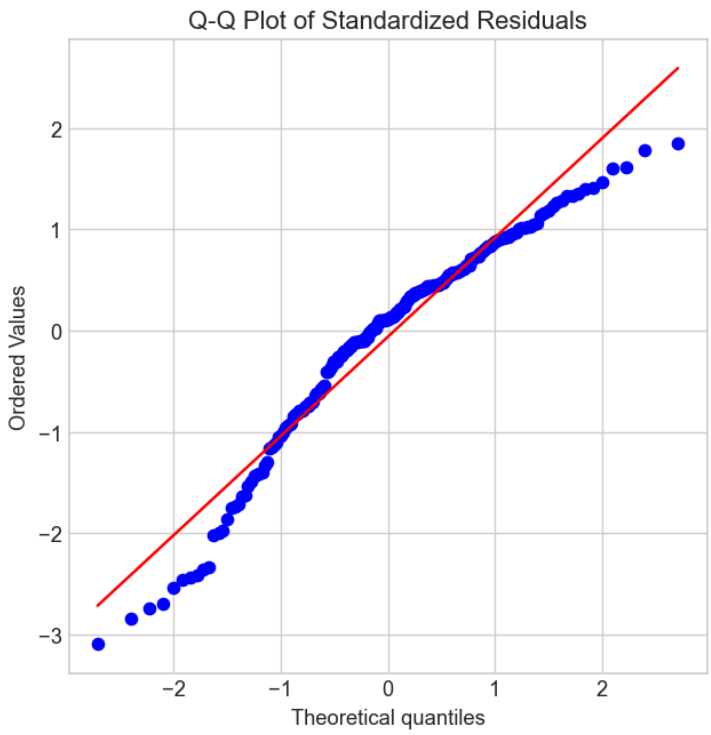

Figure 23: z_t Q-Q plot

Close to diagonal, mild tail deviations. Acceptable.

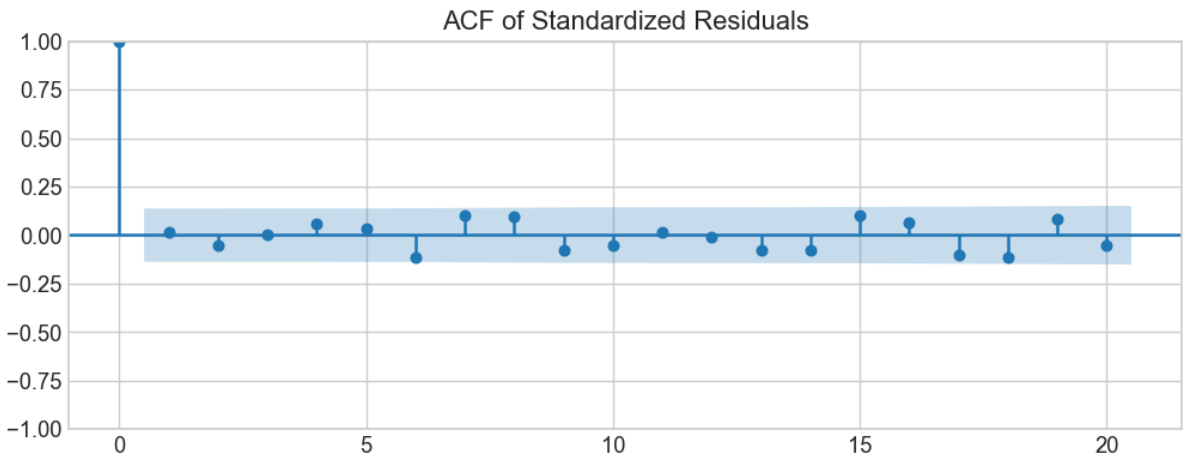

Figure 24: ACF of z_t

All lags within bands. No serial correlation in z_t.

9.3 ACF of Squared Standardized Residuals

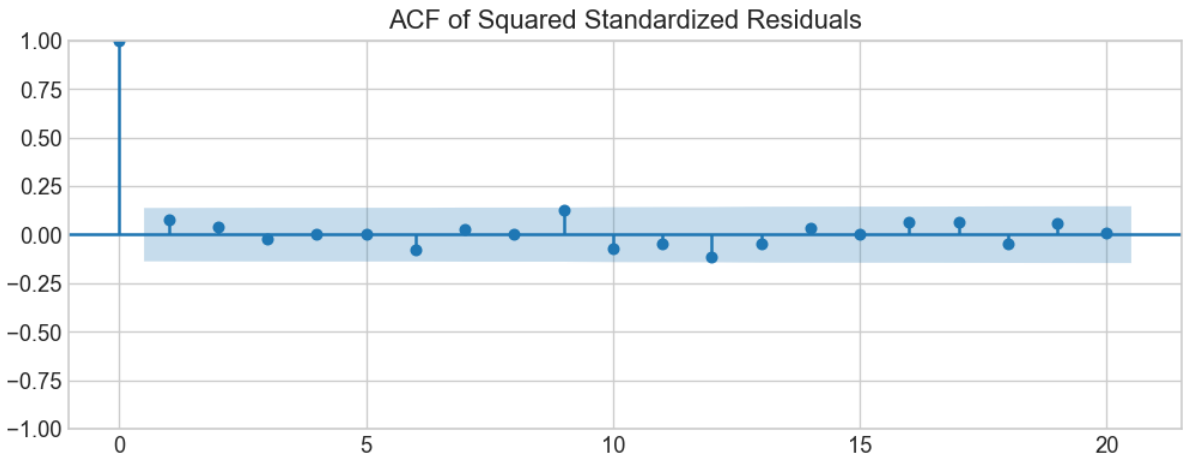

Figure 25: ACF of z_t^2

All lags inside band. No remaining ARCH effects. Model is adequate.

9.4 Ljung-Box Tests

Table 9: Ljung-Box on z_t

Lag	LB stat	p-value	Result
5	1.77	0.88	OK
10	10.70	0.38	OK
15	15.31	0.43	OK
20	23.64	0.26	OK

Table 10: Ljung-Box on z_t^2

Lag	LB stat	p-value	Result
5	1.59	0.90	OK
10	7.48	0.68	OK
15	11.56	0.71	OK
20	14.56	0.80	OK

All $p > 0.05$. No serial correlation in z_t, no ARCH effects in z_t^2. Model adequate.

10 Additional Analysis

10.1 Rolling vs Fitted Volatility

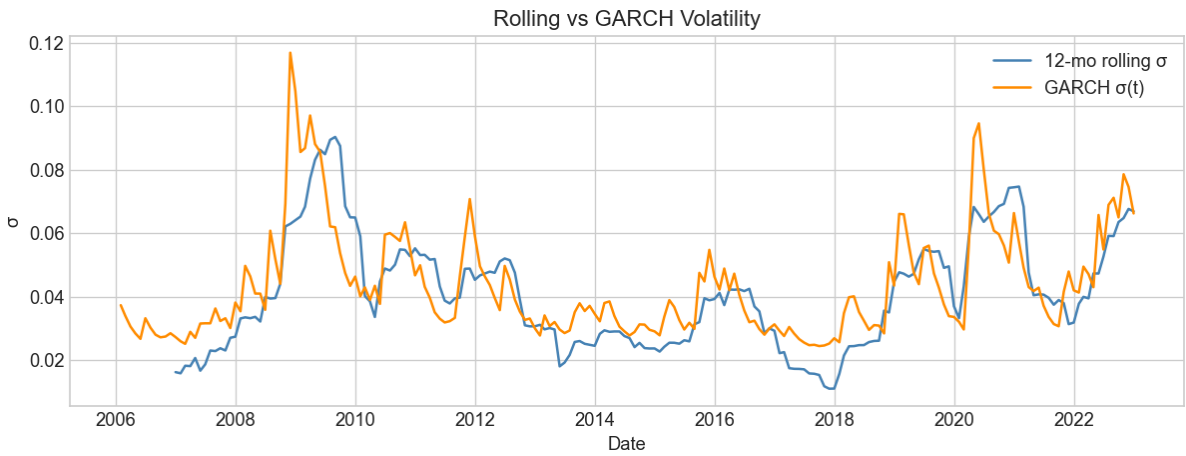

Figure 26: Rolling σ vs GARCH $\hat{\sigma}_t$

GARCH reacts faster (updates from ε_{t-1}^2, σ_{t-1}^2). Rolling estimate is smoother but lagged.

10.2 Volatility Forecast

$$\hat{\sigma}_{t+h}^2 \;\to\; \bar{\sigma}^2 = \frac{\omega}{1-\alpha_1-\beta_1} \quad \text{as } h \to \infty$$

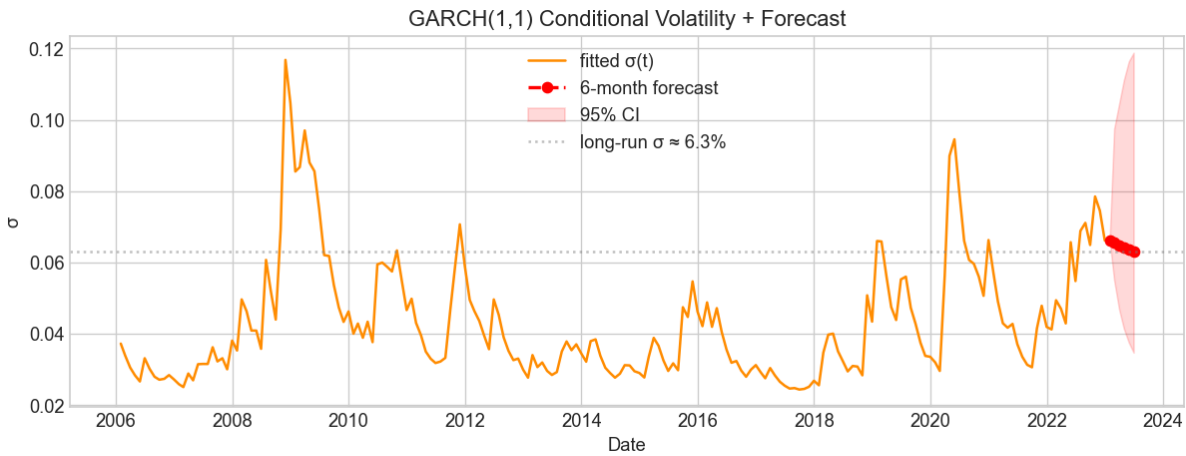

Figure 27: GARCH(1,1): fitted $\hat{\sigma}_t$ + 6-month forecast with 95% CI

Forecast (red dashed) continues from the fitted $\hat{\sigma}_t$ (orange). The 95% CI (shaded band, from 10,000 simulations) widens at longer horizons. Forecast declines toward long-run $\bar{\sigma} \approx 5.4\%$. Persistence $= 0.934$, convergence takes several months.

11 Conclusions and Limitations

11.1 Findings

1. **Stationarity**: ADF and KPSS both confirm $I(0)$, $p < 0.05$ and $p > 0.05$ respectively.
2. **Mean model**: ARIMA(0,0,0), $\hat{\mu} = 0.0071$, $p = 0.041 < 0.05$.
3. **ARCH effects**: all lags $p < 0.05$. Conditional heteroskedasticity present.
4. **Volatility model**: GARCH(1,1) by BIC. $\alpha_1 + \beta_1 = 0.934$, half-life ≈ 10.1 months.
5. **Diagnostics**: all Ljung-Box pass on z_t and z_t^2. Model adequate.

11.2 Limitations

- Still monthly data. Daily ($\sim$4000 obs) would give tighter CIs.
- Normal errors assumed but JB rejected (kurtosis = 4.46). Student-t would be better.
- No EGARCH/GJR-GARCH tested for leverage effect.
- Ljung-Box borderline at longer lags in mean model residuals.

11.3 Future Work

- Fit EGARCH / GJR-GARCH for the leverage effect
- Use Student-t error distribution
- Redo on daily data for tighter inference
- Rolling out-of-sample VaR backtesting

References

[1] Tsay, R. (2010). *Analysis of Financial Time Series.* 3rd ed., Wiley.

[2] Box, G., Jenkins, G., Reinsel, G., & Ljung, G. (2015). *Time Series Analysis: Forecasting and Control.* 5th ed., Wiley.

[3] Brockwell, P. J. & Davis, R. A. (2016). *Introduction to Time Series and Forecasting.* 3rd ed., Springer.

A Software

Python 3 with: `pandas`, `numpy`, `matplotlib`, `scipy`, `statsmodels`, `arch`.

Analysis split across 4 Jupyter notebooks:

1. `data-download.ipynb`: download SPY data
2. `eda.ipynb`: preprocessing, EDA, stationarity tests
3. `mean-modeling.ipynb`: ARIMA selection, diagnostics, ARCH test
4. `volatility-modeling.ipynb`: GARCH selection, diagnostics, forecast

Run in order. Plots saved to `plots/`.